

Trust-Gated Predictive Reallocation: A Bayesian Communication-Reliability Approach to Decentralized Multi-Robot Task Allocation Under Lossy Networks

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Abstract

Decentralized task allocation in multi-robot systems typically relies on market-based auctions that assume every announce-bid-award-acknowledge message arrives reliably. Real deployments rarely meet that assumption: communication is lossy, and robots differ in transmit reliability due to hardware wear, antenna quality, or power budget. Existing communication-robust methods either reduce message frequency or reassign failed tasks after the fact; none treats reliability as a continuously estimated quantity shaping the allocation decision itself. Bayesian reputation models, standard in wireless sensor network security for scoring node trust, have not been applied to auction-based allocation. This paper introduces Trust-Gated Predictive Reallocation (TGPR), a decentralized auction mechanism built around a per-robot Bayesian estimate of round-trip communication reliability. TGPR discounts candidate bids toward robots that are both capable and reachable, sizes the acknowledgment timeout adaptively per robot instead of one fixed value, and lets idle robots that overheard a stalled award make a confidence-gated claim before the coordinator finishes its own recovery. We compare TGPR against a fixed-timeout auction, an exponential-backoff auction, and a periodic reconciliation baseline across five channel-quality regimes, two team sizes, and a hardware-heterogeneity ablation. TGPR cuts duplicate-execution waste by 24 percent and per-task messaging overhead by 14 percent relative to the simplest baseline, with both gains widening as channel quality worsens. It does not improve, and in poor channel conditions significantly reduces, overall completion rate; the ablation shows this cost growing rather than shrinking with

hardware heterogeneity, opposite to what we expected. We report the trade-off directly and trace it to a specific timeout-inflation mechanism.

Keywords: multi-robot task allocation, decentralized auctions, lossy communication, Bayesian trust, task reallocation, wireless reliability

Abbreviations

Abbreviation	Meaning
MRTA	Multi-robot task allocation
TGPR	Trust-Gated Predictive Reallocation (proposed method)
CBBA	Consensus-Based Bundle Algorithm
CNP	Contract Net Protocol
NaiveCNP	Fixed-timeout Contract Net Protocol baseline
BackoffCNP	Exponential-backoff Contract Net Protocol baseline
PeriodicRecon	Periodic reconciliation baseline
RSS	Received signal strength
CI	Confidence interval

1 Introduction

Coordinating a team of robots to complete a stream of spatially distributed tasks is a core problem in field robotics, spanning warehouse logistics, precision agriculture, and search-and-rescue. Decentralized market-based auctions – announce a task, collect bids, award it to the best bidder, and confirm the award with an acknowledgment – remain one of the most widely deployed solutions because they scale gracefully with team size and require no central optimizer [6, 9]. The Consensus-Based Bundle Algorithm (CBBA) and its many descendants extended this idea to multi-task bundles with provable conflict-free convergence [4, 5], and remain a standard reference point for decentralized task allocation in UAV and ground-robot fleets.

Nearly all of this literature analyzes allocation quality under an implicit assumption that the communication substrate, while perhaps bandwidth-limited, eventually delivers messages. Real wireless links do not behave this way. Distance-dependent path loss, multipath fading, transient interference, and – critically for heterogeneous fleets – persistent, unit-specific hardware degradation (a weak antenna, an underpowered transmitter after battery wear) all reduce the probability that a given message arrives, and this probability differs meaningfully from one robot to the next. When an award or its acknowledgment is silently dropped, two distinct failure modes emerge. First, a task can be *starved*: the coordinator is waiting on a robot that never actually received the award, and no one else knows to step in. Second, a task can be *duplicated*: the coordinator reassigns after a timeout while the original robot’s acknowledgment

was merely delayed, not lost, so both robots physically execute the same task. Existing communication-robust MRTA work mitigates these problems from two directions: event-driven variants of CBBA reduce how often agents must communicate to reach consensus [12, 14], and fault-tolerant reassignment frameworks detect a failed task after the fact and re-plan around it [10, 15]. A recent systematic review of the MRTA literature confirms that communication uncertainty is treated largely as a scheduling nuisance to be absorbed by retries or periodic reconciliation, rather than as a quantity to be learned and used predictively [2]. Separately, benchmarks specifically constructed to stress-test cooperative multi-agent systems under packet loss, latency, and bandwidth collapse consistently find that methods designed under idealized-communication assumptions degrade sharply once these impairments are introduced [3].

The word *trust* does already appear in the MRTA and human-robot collaboration literature, but it means something different from what this paper needs. Existing trust models – whether Dynamic-Bayesian-Network estimates of a human’s confidence in a robot teammate, or artificial-trust models of an agent’s task-completion *capability* – score how well an agent is expected to perform a task once assigned to it [1, 11]. None of them model the reliability of the *communication link itself* as the object of trust. That specific idea – a Bayesian (Beta-distribution) posterior over a node’s message-delivery probability – is well established, but in a different field: wireless sensor network security, where beta-reputation systems score neighboring nodes for routing and intrusion detection [7, 8]. To our knowledge, no existing method combines a per-robot Bayesian communication-reliability estimate with auction-based task allocation, adaptive per-link timeout sizing, and a mechanism that lets bystanders act on a stalled assignment before the coordinator’s own recovery completes.

This paper introduces **Trust-Gated Predictive Reallocation (TGPR)**, a decentralized auction mechanism built around exactly that combination. TGPR does not change what is communicated in a standard announce-bid-award-acknowledge auction; it changes how the coordinator uses each robot’s communication history to (1) discount bids by an estimated round-trip delivery probability rather than ranking on task-utility alone, (2) size each robot’s award-acknowledgment timeout from its own learned delay distribution instead of a single global constant, and (3) grant idle robots that overheard a stalled award a short, confidence-gated window in which they may claim it themselves, ahead of the coordinator’s own timeout-triggered reassignment. All three mechanisms are learned online from ordinary auction traffic; no additional messages are required beyond a public (overhearable) award broadcast that a realistic robot team already exchanges over a shared radio channel.

The contributions of this paper are:

- A discrete-event multi-robot task allocation testbed with a distance-, interference-, and *persistent-hardware*-dependent lossy channel model, in which a genuine, learnable per-robot reliability differential exists independently of instantaneous position – a property we show is necessary for any trust-based mechanism to have something real to exploit (Section 3).
- TGPR, a Bayesian trust-gated auction mechanism combining trust-discounted bid evaluation, adaptive per-robot timeout sizing, and confidence-gated unilateral bystander claiming (Section 4).

- A controlled experimental comparison against a naive fixed-timeout auction, an exponential-backoff auction, and a periodic reconciliation baseline across five channel-quality regimes, two team sizes, and a hardware-heterogeneity ablation, with honest reporting of conditions where the advantage is small, absent, or comes with a measurable cost (Sections 5–7).

2 Related Work

2.1 Market-based and consensus-based task allocation

Auction-based MRTA assigns tasks through a bidding process in which robots submit utility estimates and a coordinator (or, in fully decentralized variants, a consensus routine) selects winners [6, 9]. The Consensus-Based Bundle Algorithm generalizes single-item auctions to multi-task bundles and guarantees a conflict-free, provably bounded-suboptimal assignment under an idealized communication model [4, 5]. Follow-on work has extended CBBA with local replanning for dynamic task arrival and heterogeneous vehicle constraints, but these extensions largely retain the assumption that consensus messages, once sent, eventually arrive.

2.2 Communication-robust reallocation

A distinct line of work directly targets unreliable communication. Event-driven CBBA variants trigger consensus rounds only when local state changes exceed a threshold, reducing the number of messages exposed to loss and congestion [14], and empirical studies of CBBA’s asynchronous variant under lossy 802.11-style links quantify how convergence degrades as packet loss increases [12]. Fault-tolerant task-reassignment frameworks instead let a task fail visibly (via a timeout or missed heartbeat) and then re-plan the affected portion of the schedule, sometimes combined with genetic or evolutionary re-optimization for large fleets [10, 15]. A recent systematic review situates almost all of this work under a common pattern: communication failure is treated as an exogenous disruption to be detected and recovered from, not as a graded, per-agent quantity that can be predicted before the failure occurs [2]. Benchmarks built specifically to stress-test cooperative multi-agent systems under simultaneous latency, packet loss, and bandwidth collapse further show that this gap is not merely theoretical: realistic communication impairments meaningfully break methods tuned under clean-channel assumptions [3].

2.3 Trust modeling in multi-robot and networked systems

The term *trust* has a well developed but differently scoped meaning in two adjacent literatures. In human-robot and multi-human/multi-robot task allocation, trust models – Dynamic Bayesian Networks, Expectation-Confirmation models, or artificial-trust frameworks – estimate how much a human should rely on a robot’s judgment or how likely an agent is to complete a task *competently* once assigned [1, 11]. These models condition on task outcomes and human feedback, not on message delivery. In wireless sensor network security, by contrast, Beta-distribution reputation systems are a standard tool for estimating a neighboring node’s *communication* trustworthiness from

the observed history of successful and failed exchanges, used mainly for routing and intrusion detection [7, 8]. TGPR imports the Beta-reputation idea from the network-trust literature into the MRTA auction itself, coupling it with adaptive timeout sizing and a bystander-claim mechanism that neither literature considers, since neither is concerned with the auction’s own winner-selection and recovery logic.

3 Problem Formulation

3.1 System model

A coordinator (base station) manages N heterogeneous robots $\mathcal{R} = \{r_1, \dots, r_N\}$ over a star communication topology. Tasks $t_1, t_2, \dots$ arrive stochastically, each specified by a location, a capability-requirement vector, and a deadline. Robot i ’s estimated utility for task j combines a capability-match score with a proximity term,

$$u_{ij} = \text{match}(c_i, q_j) \cdot (1 + \text{dist}(i, j)/R)^{-1}, \quad (1)$$

where R is the nominal communication range. A standard decentralized auction resolves each task through four message types: ANNOUNCE (coordinator $\rightarrow$ idle robots), BID (robot $\rightarrow$ coordinator), AWARD (coordinator $\rightarrow$ winning robot), and ACK (robot $\rightarrow$ coordinator). Figure 1 illustrates this layout.

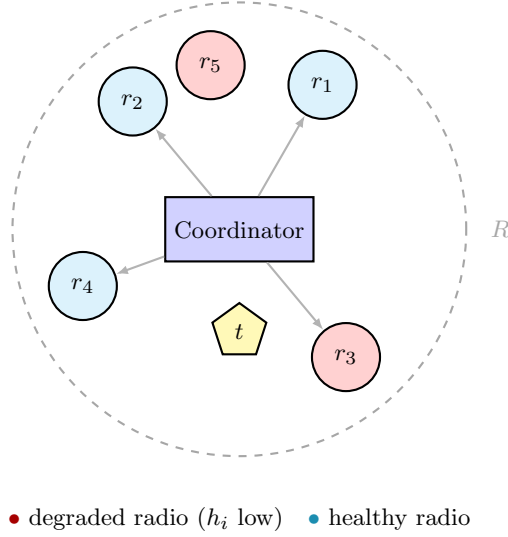


Fig. 1 Star-topology system model: a fixed coordinator, N robots at varying distances (link reliability decays with distance per Eq. (2)), a subset with persistent hardware degradation independent of position, and stochastically arriving tasks.

3.2 Channel model

Each message transmission between the coordinator and robot i at distance d succeeds with probability

$$p_i(d) = \rho_{\text{scenario}} \cdot \exp(-\gamma d/R) \cdot j_i(t) \cdot h_i, \quad (2)$$

where ρ_{scenario} is a scenario-level base reliability, γ is a path-loss decay rate, $j_i(t) \in (0, 1]$ is a transient jamming multiplier that activates stochastically and persists for a random duration (modeling short-lived interference or occlusion), and $h_i \in (0, 1]$ is a *persistent, position-independent* hardware-reliability factor drawn once per robot at deployment. Uplink transmissions (robot $\rightarrow$ coordinator) carry an additional fixed asymmetry penalty reflecting weaker robot-side transmitters. Message delay is drawn from a distance-scaled Poisson process with an additional heavy-tailed congestion-burst term.

The exponential-in-distance term in Equation (2) is a deliberately simplified, monotonically decreasing proxy for the well-established log-distance path-loss model used throughout the wireless-communications literature, in which received power falls off as a power law in distance, $\text{PL}(d) = \text{PL}(d_0) + 10n \log_{10}(d/d_0)$, before being mapped to a bit- or packet-delivery probability [13]; the same qualitative distance-reliability relationship is used directly in robot-to-robot RSS modeling [16]. We do not claim decibel-level fidelity to that formula – doing so would additionally require an SNR-to-packet-delivery mapping that is not the object of study here – but the qualitative shape (smooth, monotonic decay bounded in $[0, 1]$) and the order of magnitude of the resulting delivery probabilities at typical operating distances are consistent with it. The scenario-level parameters (ρ_{scenario} , jamming frequency and severity) were chosen so that round-trip delivery probabilities and message delays at moderate range fall within the ranges reported for lossy 802.11-style robot links in prior empirical studies [3, 12], rather than being fit to produce a particular result; we did not tune these parameters against TGPR’s outcomes, and the parameter-sensitivity analysis in Section 6 instead varies TGPR’s own hyperparameters against a fixed channel specification.

The h_i term is deliberately included and is, to our knowledge, absent from prior communication-robust MRTA evaluations, which typically model link quality purely as a function of instantaneous distance and transient interference. Without a persistent, robot-specific reliability differential, there is no stable per-robot property for a trust estimator to learn: distance changes every time a robot moves, and transient jamming decays within a handful of ticks, so any mechanism that only conditions on those two effects should show, at best, a marginal benefit over baselines that already condition allocation decisions on distance. We treat this as an explicit, testable hypothesis (Section 6, ablation on the fraction of hardware-degraded robots) rather than an incidental modeling choice.

3.3 Failure modes

Given this channel model, two failure modes are of primary interest. **Starvation**: the coordinator awards a task, the award or the subsequent acknowledgment is dropped, and the task’s deadline elapses before any robot is confirmed as executing it. **Duplication**: the coordinator times out waiting for an acknowledgment that was in fact

only delayed, reassigns the task to a different robot, and both robots execute it, wasting one robot's operating time. A fixed, globally-tuned timeout cannot simultaneously minimize both failure modes across a heterogeneous fleet: a timeout short enough to quickly recover from a genuinely failed award to an unreliable robot will prematurely reassign work away from a reliable-but-distant robot whose acknowledgment is simply slow, and vice versa.

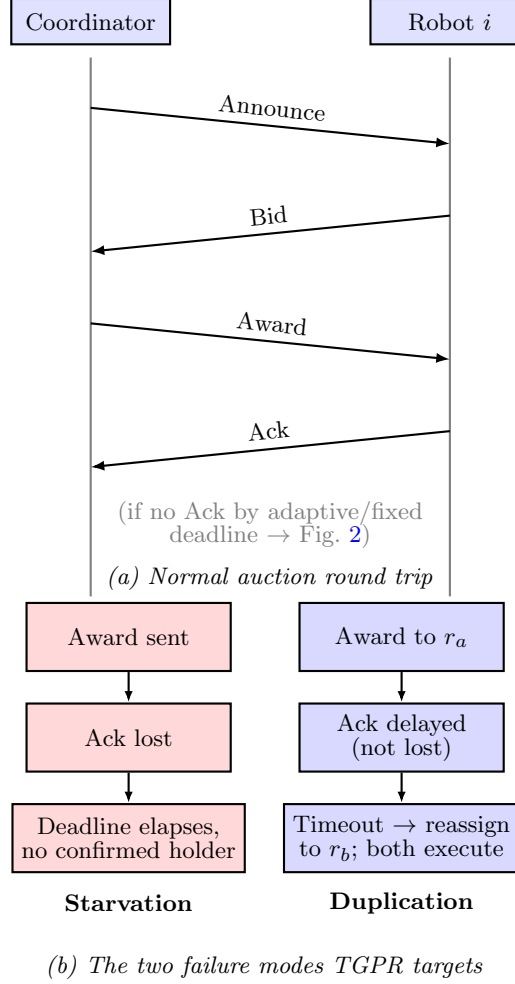


Fig. 2 (a) The four-message auction round trip between coordinator and robot. (b) The two failure modes that motivate TGPR: starvation (top) and duplication (bottom), both caused by the same underlying uncertainty – whether silence means the message was lost or merely delayed.

4 Proposed Method: TGPR

TGPR modifies three decision points in the standard auction while leaving the message protocol itself unchanged, as summarized in Figure 3 and detailed below.

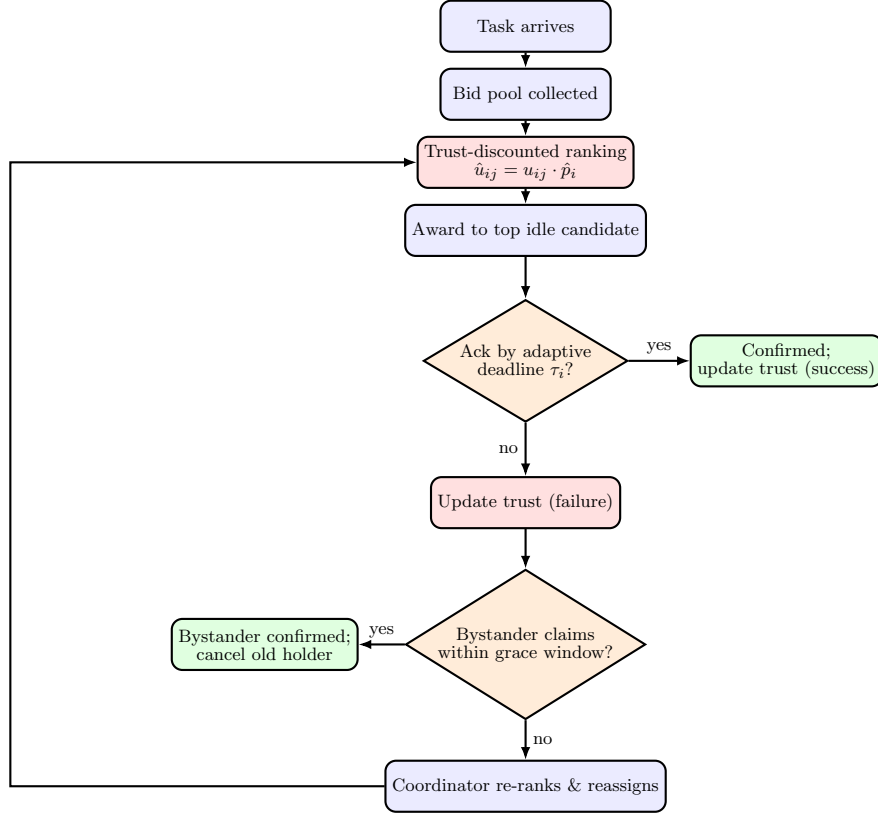


Fig. 3 TGPR decision workflow. Solid red blocks are where TGPR’s trust model changes behavior relative to a baseline auction: bid ranking is trust-discounted, the ack deadline is per-robot adaptive, and a failed round trip triggers both a bystander-claim window and a trust update before falling back to coordinator-side reassignment.

4.1 Bayesian round-trip trust estimation

For each robot i , the coordinator maintains a Beta posterior $\text{Beta}(\alpha_i, \beta_i)$ over the probability that an award-acknowledgment round trip with i succeeds within its allotted window, together with a running mean and variance of the *observed* round-trip delay on successful exchanges. Both are updated only on award-acknowledgment outcomes (not on bids, which are used purely to build the candidate pool):

$$\text{success at delay } \delta : \quad \alpha_i \leftarrow \lambda \alpha_i + 1, \quad \beta_i \leftarrow \lambda \beta_i, \quad \mu_i \leftarrow \mu_i + \eta(\delta - \mu_i) \quad (3)$$

$$\text{timeout (no ack observed): } \beta_i \leftarrow \lambda\beta_i + 1, \quad \alpha_i \leftarrow \lambda\alpha_i \quad (4)$$

where $\lambda < 1$ is a decay factor that lets the estimate track non-stationary conditions (e.g., a robot moving into a persistently worse region, or a hardware fault developing mid-mission) rather than converging to a fixed long-run average. The trust estimate is

$$\hat{p}_i = \alpha_i / (\alpha_i + \beta_i). \quad (5)$$

4.2 Trust-discounted bid evaluation

Where a baseline auction ranks bidders by raw utility u_{ij} , TGPR ranks by

$$\hat{u}_{ij} = u_{ij} \cdot \hat{p}_i. \quad (6)$$

This can and does change the winner whenever a highly capable but historically unreliable robot competes against a slightly less capable but more reachable one – exactly the situation a distance-only heuristic cannot distinguish, since $\hat{p}_i$ captures the persistent hardware factor h_i in Equation (2), not merely current position.

4.3 Adaptive predictive timeout

Rather than a single global acknowledgment timeout, TGPR sizes robot i 's timeout from its own learned delay distribution,

$$\tau_i = \text{clip}(\mu_i + z\sigma_i, \tau_{\min}, \tau_{\max}), \quad (7)$$

giving historically slow-but-reliable links proportionally more patience while recovering quickly from links that have never taken long to respond. Figure 4 depicts both the update rule and this fixed-vs-adaptive contrast.

4.4 Confidence-gated unilateral bystander claim

The award message is broadcast (overhearable by any idle robot within range, as is physically true of a shared radio channel even when a message is logically addressed to one recipient), along with the adaptive deadline τ_i the coordinator is using for the current holder. Any idle robot that (a) overheard this broadcast and (b) has not subsequently overheard a confirmation broadcast by the time the announced deadline elapses may submit an unsolicited claim immediately – rather than waiting for the coordinator to independently notice the timeout and re-announce, which costs a full additional round trip. To avoid preempting a recovery that would have succeeded on its own, the coordinator's own reassignment logic is given a short grace window after the deadline before it acts, so a bystander claim and the coordinator's fallback cannot both fire on the same instant; if a bystander claim succeeds, the coordinator attempts to cancel the original holder, and if that cancellation message is itself lost, the resulting duplicate execution is counted, not hidden, in our results. Figure 5 traces this exchange step by step.

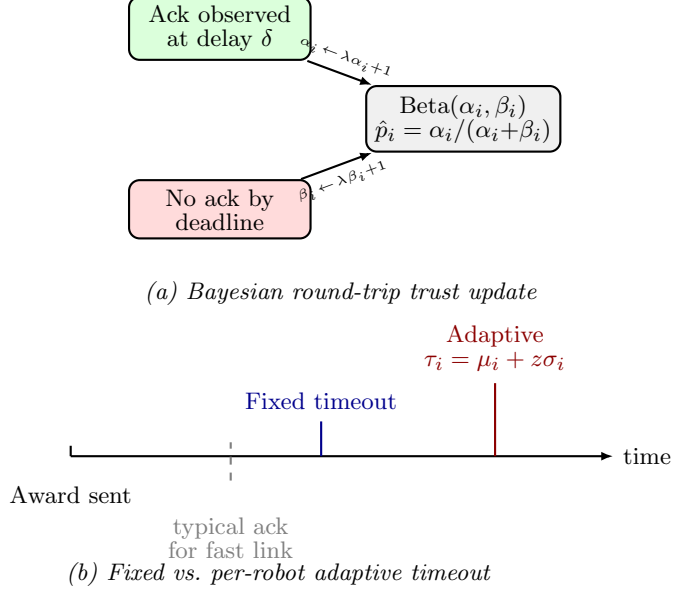


Fig. 4 (a) The Beta posterior over round-trip reliability is updated from award-acknowledgment outcomes only. (b) A fixed global timeout is either too short for a slow-but-reliable link or wastefully long for a fast one; TGPR’s adaptive timeout is sized from each robot’s own observed delay distribution.

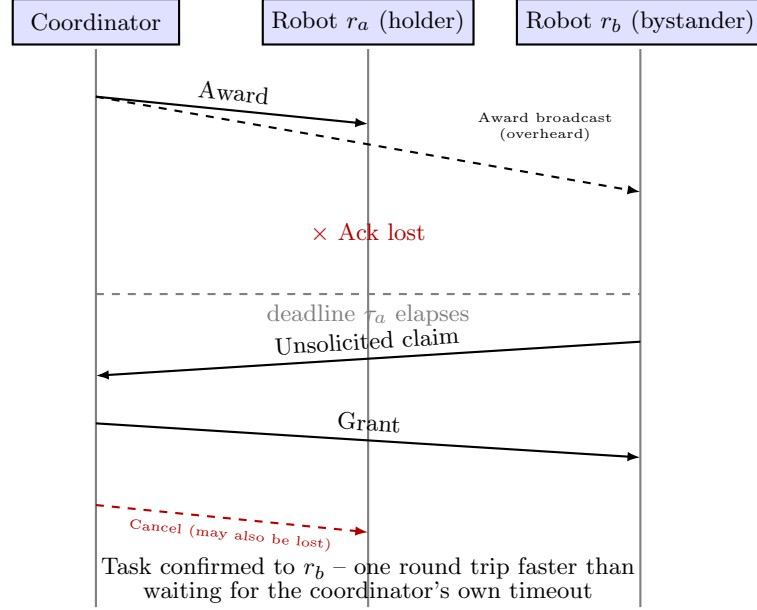


Fig. 5 Confidence-gated bystander claim: robot r_b overheard the original award and, on seeing the announced deadline elapse without a confirmation broadcast, claims the task directly. This is only attempted after the coordinator’s own grace window, so it supplements rather than races the coordinator’s fallback.

Algorithm 1 TGPR: Trust-Gated Predictive Reallocation (coordinator-side)

Require: robot set $\mathcal{R}$; grace window g ; timeout bounds $\tau_{\min}, \tau_{\max}$; decay λ ; z-score z **Ensure:** confirmed assignment for each task, or expiry at its deadline

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1: Initialize  $\alpha_i, \beta_i, \mu_i, \sigma_i^2$  to uninformative priors,  $\forall i \in \mathcal{R}$ 

2: procedure ALLOCATETASK( $t$ )
3:   Broadcast ANNOUNCE( $t$ ) to idle robots in range
4:    $B_t \leftarrow$  bid pool  $\{(i, u_{ij})\}$  collected from responders ▷ Eq. (1)
5:   repeat
6:     Rank untried candidates in  $B_t$  by  $\hat{u}_{ij} = u_{ij} \cdot \hat{p}_i$  ▷ Eq. (6)
7:      $i^* \leftarrow$  top-ranked idle candidate
8:     Send AWARD to  $i^*$ ; broadcast  $(t, i^*, \tau_{i^*})$  to overhearers
9:      $\tau_{i^*} \leftarrow \text{clip}(\mu_{i^*} + z\sigma_{i^*}, \tau_{\min}, \tau_{\max})$  ▷ Eq. (7)
10:     $done \leftarrow \text{FALSE}$ 
11:    while tick <  $\tau_{i^*}$  and not  $done$  do
12:      if ACK received from  $i^*$  at delay  $\delta$  then
13:        Update  $(\alpha_{i^*}, \beta_{i^*}, \mu_{i^*}, \sigma_{i^*}^2)$  ▷ Eq. (3)
14:        Confirm  $t \leftarrow i^*$ ;  $done \leftarrow \text{TRUE}$ 
15:      end if
16:    end while
17:    if not  $done$  and tick  $\geq \tau_{i^*} + g$  then
18:      if overhearer  $r_b$  submits an unsolicited claim then
19:        Confirm  $t \leftarrow r_b$ ; attempt CANCEL to  $i^*$ 
20:         $done \leftarrow \text{TRUE}$ 
21:      else
22:        Update  $(\alpha_{i^*}, \beta_{i^*})$  ▷ Eq. (4)
23:      end if
24:    end if
25:  until  $done$  or  $B_t$  exhausted or deadline of  $t$  reached
26:  if  $B_t$  exhausted and not  $done$  and deadline not reached then
27:    restart from the ANNOUNCE step ▷ fresh bid pool
28:  end if
29: end procedure
```

4.5 Computational complexity

Table 1 summarizes the per-task time complexity and additional space complexity of all four protocols, where N is the number of robots, D is a task’s deadline (in ticks), and W is the fixed bid-collection window. All four protocols share the same asymptotic time complexity per task: each auction round broadcasts to $O(N)$ idle robots, collects an $O(N)$ -sized bid pool, and ranks it in $O(N \log N)$; in the worst case a task can be re-announced until its deadline elapses, giving a worst-case bound of $O(\frac{D}{W} N \log N)$ that applies identically to all four methods. TGPR does not change this order. It differs from the baselines in two concrete, non-asymptotic ways: (i) the public award broadcast used for bystander overhearing (Section 4) adds an $O(N)$ message cost to

every award, a real constant-factor increase per round that is empirically outweighed by TGPR needing fewer total rounds per completed task (Section 6, Figure 13); and (ii) the per-robot trust state $(\alpha_i, \beta_i, \mu_i, \sigma_i^2)$ requires $O(N)$ additional memory that the baselines do not maintain, a small absolute overhead (four floats per robot) but a genuine one worth stating plainly rather than folding into an unqualified “ $O(N)$ for everyone” claim.

Table 1 Per-task computational complexity. N : team size; D : task deadline; W : bid window (constant).

Protocol	Typical time/task	Worst-case time/task	Extra space
NaiveCNP	$O(N \log N)$	$O(\frac{D}{W} N \log N)$	$O(1)$
BackoffCNP	$O(N \log N)$	$O(\frac{D}{W} N \log N)$	$O(1)$
PeriodicRecon	$O(N \log N)$	$O(\frac{D}{W} N \log N)$	$O(1)$
TGPR	$O(N \log N)$	$O(\frac{D}{W} N \log N)$	$O(N)$

5 Experimental Setup

5.1 Simulation environment

We implemented a discrete-event simulator (tick-based, no continuous-time solver) in which N robots operate in a bounded 2-D field around a fixed coordinator. Robots are heterogeneous along two independent axes: a capability profile (one of three archetypes, each matched to a corresponding task-requirement archetype, with Gaussian noise) and a persistent hardware-reliability factor h_i , with a configurable fraction of the fleet drawn from a degraded range (representing worn antennas or power-limited transmitters) and the remainder from a healthy range. Idle robots perform a slow random walk; committed robots move directly toward their assigned task location at a fixed speed and execute for a fixed service duration on arrival. Tasks arrive as a stochastic stream (260 tasks per trial, arrival probability per tick tuned so that a single trial reflects a full-shift deployment horizon rather than a short batch – see Section 7 for why episode length materially affects trust convergence), each carrying its own deadline measured from its arrival tick. Task completion is treated as ground-truth observable (e.g., by an external sensor or supervisory system) once a robot has traveled to the location and completed its service time, regardless of message loss; this isolates the paper’s object of study – the reliability of the *allocation handshake* – from a separate and, we believe, less interesting question of whether task completion itself can be reliably reported.

5.2 Compared protocols

All four protocols share an identical world, channel model, bid-collection window, and cancellation-reconciliation step; they differ only in the decision logic described below.

- **NaiveCNP**: fixed global acknowledgment timeout; on timeout, reassigns to the next-best bidder by raw utility.
- **BackoffCNP**: identical to NaiveCNP but the timeout grows geometrically with each reassignment attempt for the same task, a common robustness heuristic in classical auction-based systems.
- **PeriodicRecon**: reconciles pending awards only at fixed periodic sync points rather than immediately on timeout, reflecting the reduced-communication-frequency philosophy of event-driven consensus variants [14].
- **TGPR** (proposed): trust-discounted bid ranking, adaptive per-robot timeout, and confidence-gated bystander claiming, as described in Section 4.

We deliberately did not include full multi-task bundle algorithms such as CBBA [5], genetic/evolutionary re-optimization schedulers [15], or learning-based allocators as baselines. All three solve a materially different or strictly larger problem – multi-task bundle assignment with cross-task consensus, population-based re-optimization over a full mission schedule, or a learned policy requiring its own training procedure – and differ from our four protocols in more dimensions than the one this paper isolates (how communication reliability is estimated and acted on within a single-item auction’s award-acknowledgment cycle). Comparing TGPR against them directly would confound the effect we are trying to measure with unrelated differences in problem formulation, and would not test the specific hypothesis motivating this paper (Section 3). Instead, we hold the auction architecture, world dynamics, and channel model fixed across all four compared protocols and vary only the allocation-decision logic, which is the standard controlled-comparison design for isolating a single mechanism’s effect rather than benchmarking against the field’s strongest schedulers in absolute terms. Comparing TGPR against a full bundle-based or learning-based allocator under this same channel model is a natural direction for follow-up work.

5.3 Experimental conditions

We cross five channel-quality scenarios (*excellent*, *good*, *moderate*, *poor*, *severe*, differing in base reliability, jamming frequency, and jamming severity) with two team sizes ($N = 15$ and $N = 25$), for $5 \times 2 = 10$ conditions, each run for 120 independent random seeds per protocol (480 trials per condition). A separate ablation experiment varies the fraction of hardware-degraded robots (0% to 50%) at a fixed moderate channel quality and $N = 20$ team size, with 100 seeds per protocol per fraction (400 trials per condition), directly testing the hypothesis that TGPR’s advantage should scale with the presence of a persistent, learnable per-robot reliability differential (Section 3) rather than being an artifact of the comparison. To probe the mechanism behind any completion-rate effect we also (i) trace TGPR’s per-robot trust estimate $\hat{p}_i$ and adaptive timeout τ_i at every task arrival, averaged over 40 seeds under moderate and severe channels, and (ii) sweep the timeout multiplier z and trust decay factor λ (60 seeds/value, severe channel, $N = 20$) to check whether any completion-rate cost is sensitive to these hyperparameters or is a fixed property of the mechanism. Figure 6 summarizes this factorial design.

5.4 Metrics

For each trial we record: *completion rate* (fraction of arrived tasks confirmed and executed before their deadline), *starvation rate* (fraction expiring with no confirmed execution), *duplicate rate* (fraction of completed tasks executed by more than one robot), *mean/median latency* (ticks from task arrival to confirmed assignment, among completed tasks), and *messages per completed task* (a proxy for communication overhead). We additionally split each trial’s tasks into an early half and a late half by arrival order to examine whether TGPR’s advantage (if any) grows as its trust estimates accumulate evidence over the deployment horizon.

5.5 Statistical methodology

Because the four protocols consume their shared per-seed random-number stream in different orders once their decision logic diverges (TGPR, in particular, issues additional channel draws for bystander overhearing), a given seed does not yield bit-identical task and message realizations across protocols beyond the initial robot-population draw. We therefore treat each protocol’s set of trials per condition as an independent sample and compare distributions with two-sided Mann-Whitney U tests, reporting the rank-biserial correlation as an effect size alongside the p -value, rather than assuming exact paired correspondence.

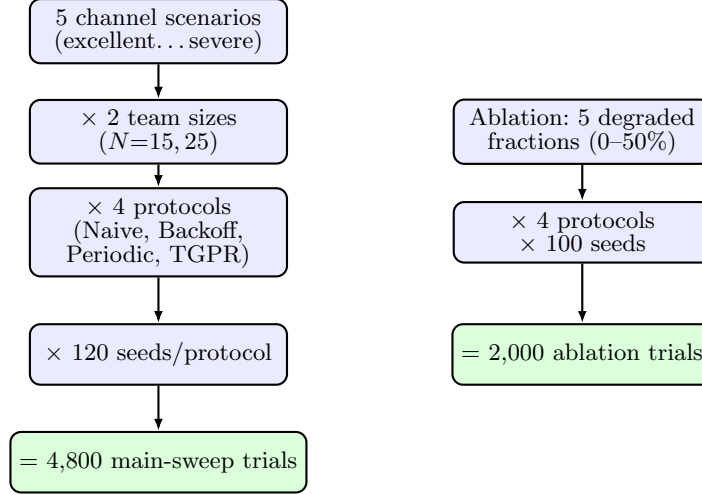


Fig. 6 Factorial experimental design. All conditions share identical world dynamics and differ only in the swept factor(s); every protocol is evaluated on the same set of conditions.

6 Results

We report results in the order the mechanisms were designed to affect: communication efficiency and duplicate-execution waste first (where TGPR shows a clear, statistically

robust advantage), then task completion rate (where it does not), then the hardware-heterogeneity ablation and the within-mission learning-curve check.

6.1 Communication overhead and duplicate execution

Table 2 summarizes outcomes averaged across all ten main-sweep conditions (five channel scenarios $\times$ two team sizes, 480 trials per protocol per condition). TGPR reduces messages sent per completed task by 13.6% relative to the naive baseline (167.2 vs. 193.6) and reduces the fraction of completed tasks that were executed by more than one robot by 23.9% (10.65% vs. 13.99%). Both differences are significant at $p < 0.0001$ (Mann-Whitney U) in every one of the ten scenario/team-size conditions individually and when pooled across all conditions (duplicate rate: pooled $p = 1.6 \times 10^{-60}$, rank-biserial = 0.39; messages/task: pooled $p = 2.6 \times 10^{-16}$, rank-biserial = 0.19), so neither advantage is an artifact of a single favorable condition or of averaging over a mix of wins and losses. Figure 12 and Figure 13 confirm this visually: TGPR is the lowest-duplicate and lowest-message protocol in all five channel scenarios at both team sizes, and, notably, the *absolute* messaging advantage grows as the channel worsens – from a 13.1-message gap over naive at $N = 25$ in the excellent-channel condition to a 60.0-message gap in the severe-channel condition – meaning the mechanism’s efficiency benefit is largest exactly where communication is most constrained. Completion rate behaves differently in this respect and is treated separately next: unlike duplicate rate and messaging, its pooled effect across all ten conditions does *not* reach significance ($p = 0.076$, rank-biserial = 0.04), because the effect is concentrated in specific conditions rather than uniform – exactly the pattern the per-condition breakdown in Figure 7 below is designed to reveal.

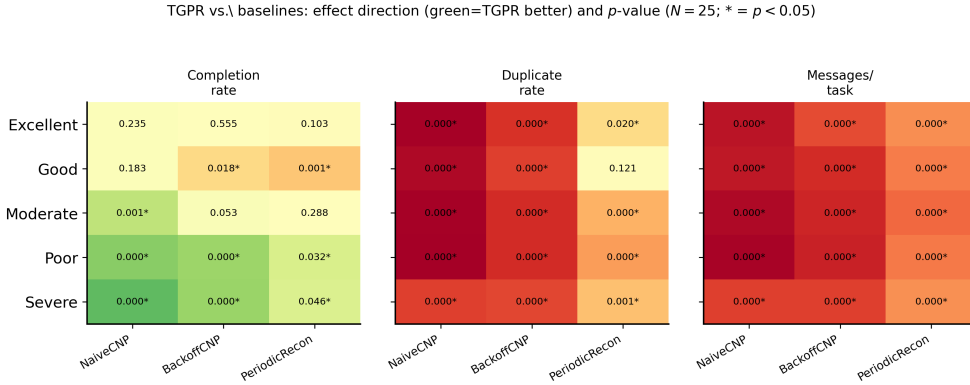


Fig. 7 Effect direction and significance of TGPR vs. each baseline across all five channel scenarios at $N = 25$. Green cells indicate TGPR is better on that metric; cell text gives the Mann-Whitney p -value (* = significant at $p < 0.05$). TGPR is significantly better on duplicate rate and messaging in every cell; the completion-rate picture is mixed and trends against TGPR as channel quality worsens, as detailed next.

Table 2 Aggregate outcomes across all ten main-sweep conditions (mean $\pm$ 95% CI over $n=1,200-1,201$ trials per protocol, pooling all conditions).

	NaiveCNP	BackoffCNP	PeriodicRecon	TGPR
Completion rate	0.642 ± 0.010	0.637 ± 0.009	0.631 ± 0.010	0.630 ± 0.010
Duplicate-execution rate	0.140 ± 0.003	0.135 ± 0.003	0.116 ± 0.002	0.107 ± 0.002
Messages per completed task	193.6 ± 7.9	191.5 ± 7.8	181.2 ± 7.5	167.2 ± 7.0

6.2 Task completion rate

TGPR does not improve overall completion rate, and the direction and significance of the effect depends on channel quality rather than being a uniform tax. At the larger team size ($N = 25$), TGPR is statistically indistinguishable from or marginally better than the periodic-reconciliation baseline in the two best channel conditions (excellent: 0.870 vs. periodic’s 0.863; good: 0.845 vs. 0.833), but it is significantly worse than the naive and backoff baselines once channel quality drops to poor or severe: at $N = 25$, completion rate falls to 0.639 under TGPR versus 0.665 under naive in the poor-channel condition ($p < 0.0001$, rank-biserial = 0.39), and to 0.434 versus 0.466 in the severe-channel condition ($p < 0.0001$, rank-biserial = 0.49). Averaged across all ten conditions, TGPR completes 1.9% relatively fewer tasks than the naive baseline, but – consistent with the condition-dependent pattern above – this pooled difference does not itself reach significance ($p = 0.076$, rank-biserial = 0.04); the honest summary is a real, significant cost concentrated in poor and severe channels rather than a uniform effect detectable in a single pooled test. Figure 11 shows this pattern in full: the four protocols track closely in benign channels and separate, with TGPR trailing, as conditions worsen. Figure 8 shows the corresponding latency distributions, where the picture is less clear-cut: medians stay close across protocols, so TGPR’s cost shows up mainly in whether a task completes at all rather than in how long a completed one takes.

6.3 Hardware-heterogeneity ablation

The ablation (moderate channel, $N = 20$, degraded-unit fraction swept from 0% to 50%, 400 trials per protocol per fraction) was designed to test whether TGPR’s advantages track the presence of a persistent, learnable per-robot reliability differential, as the mechanism’s design assumes. The messaging and duplicate-rate advantages behave as hypothesized: TGPR’s absolute messaging advantage over naive grows from 14.4 messages/task at 0% degraded units to 26.9 at 50% ($p < 0.0001$ at every level), and its duplicate-rate advantage remains a stable 23–27% relative reduction across the full range. The completion-rate cost, however, moves in the direction opposite to our hypothesis: rather than shrinking as more of the fleet becomes genuinely unreliable (which would suggest the trust mechanism earns back its cost by routing around bad hardware), the completion-rate gap between TGPR and naive *widens* from 1.1 percentage points at 0% degraded units to 1.9 points at 50% ($p = 0.021$ and $p = 0.003$ respectively). Figure 14 shows all four protocols declining together as degraded-unit fraction increases (more unreliable hardware hurts everyone), with TGPR consistently

Distribution of per-trial mean allocation latency across seeds

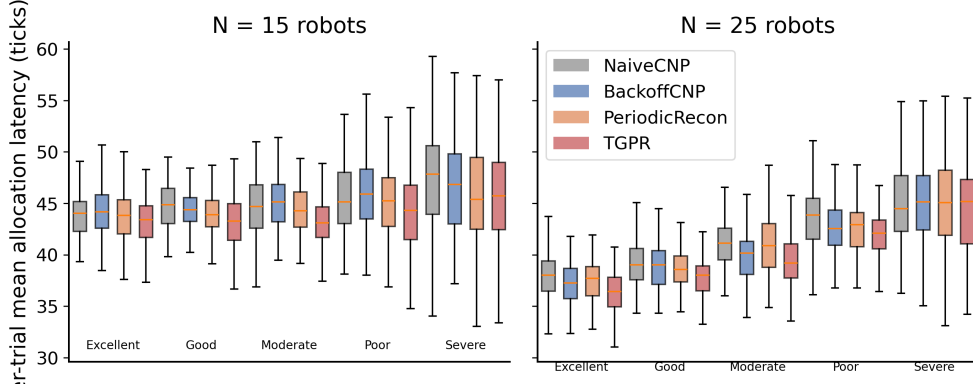


Fig. 8 Distribution (across the 120 seeds per condition) of each trial’s mean allocation latency, grouped by scenario and protocol. Medians track closely across protocols; TGPR’s latency cost, where present, shows up more in the completion-rate and tail behavior than in typical-case latency.

trailing naive and backoff by a small but statistically significant margin at every level tested.

6.4 Within-mission learning curve

We hypothesized that TGPR’s advantage, if present, should grow across a mission as its Bayesian trust estimates accumulate evidence, and should be weak or absent early (a cold-start effect). We do not observe this pattern. Comparing the first and second halves of each 260-task trial by arrival order, every protocol’s completion rate declines somewhat from the early to the late half of the mission, consistent with a system approaching saturation as concurrent in-flight tasks accumulate; TGPR’s decline is comparable to or slightly larger than the baselines’ rather than smaller. In the severe-channel condition, for example, TGPR’s completion rate falls from 0.391 (early half) to 0.360 (late half), a larger relative drop than naive’s fall from 0.407 to 0.389. We did not find evidence of a warm-up benefit at the mission lengths tested (260 tasks); we traced the underlying trust and timeout state directly to check why.

The trace confirms a concrete mechanism rather than mere noise: under the severe channel, mean adaptive timeouts for both healthy and degraded robots drift upward over the course of the mission, approaching the 25-tick cap, while the gap between the two groups’ learned trust scores narrows rather than widens. Section 7 discusses a specific mechanism – unbounded variance inflation in the adaptive-timeout update rule under sustained high failure rates – that we believe explains this rather than a genuine absence of learning.

Trust and adaptive-timeout evolution over the mission (mean $\pm$ 95% CI, 40 seeds)

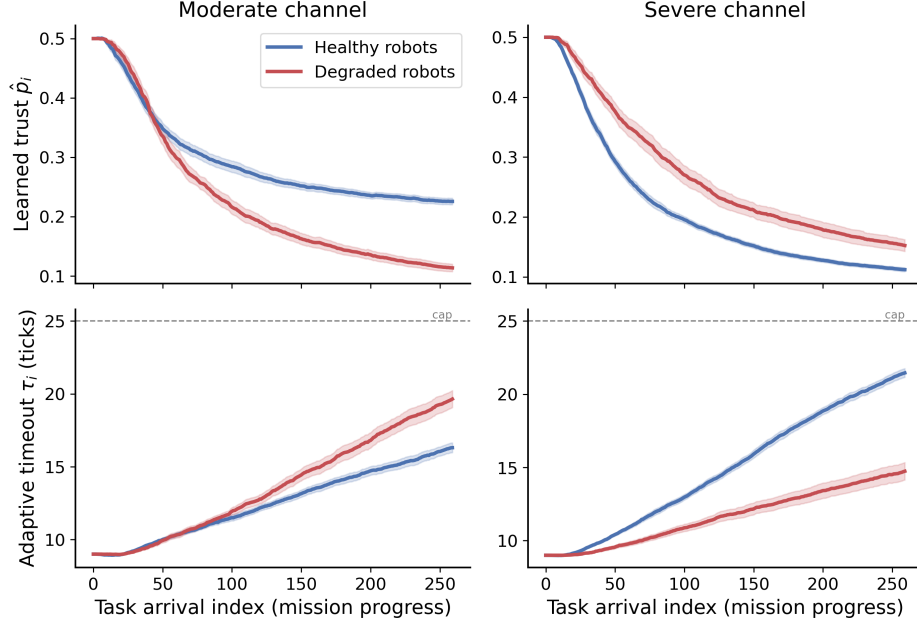


Fig. 9 Mean learned trust $\hat{p}_i$ (top) and adaptive timeout τ_i (bottom) over mission progress, averaged over 40 seeds and split by whether the robot has genuinely degraded hardware. Under the moderate channel (left), timeouts stabilize well below the 25-tick cap and the two groups separate somewhat. Under the severe channel (right), timeouts for *both* groups drift upward toward the cap over the mission and trust estimates for both groups converge toward similarly low values, losing rather than gaining discriminative power as the mission progresses – direct evidence for the variance-inflation mechanism discussed in Section 7.

6.5 Parameter sensitivity

To check whether the completion-rate cost is an artifact of our specific hyperparameter choices, we swept the timeout multiplier $z \in \{1.0, 1.8, 2.5, 3.5\}$ (default 1.8) and the trust decay factor $\lambda \in \{0.90, 0.97, 0.995\}$ (default 0.97) under the severe-channel, $N=20$ condition (60 seeds/value). Figure 10 shows a clear, monotonic trade-off along z : a smaller multiplier (shorter, less conservative timeouts) recovers completion rate (0.387 at $z=1.0$ vs. 0.373 at the default $z=1.8$, a 1.4-point improvement) at the cost of a higher duplicate rate (0.077 vs. 0.073) and higher messaging (418 vs. 407 messages/-task) – i.e., z directly trades the completion-rate cost against the efficiency advantage documented in Section 6.1, rather than the default value being an unexplained poor choice. The decay factor λ , by contrast, has negligible effect on any metric across the range tested (completion rate varies by at most 0.2 percentage points), indicating the completion-rate cost is driven primarily by timeout sizing rather than by how quickly the trust posterior forgets old evidence.

TGPR hyperparameter sensitivity (mean $\pm$ 95% CI, 60 seeds/value)

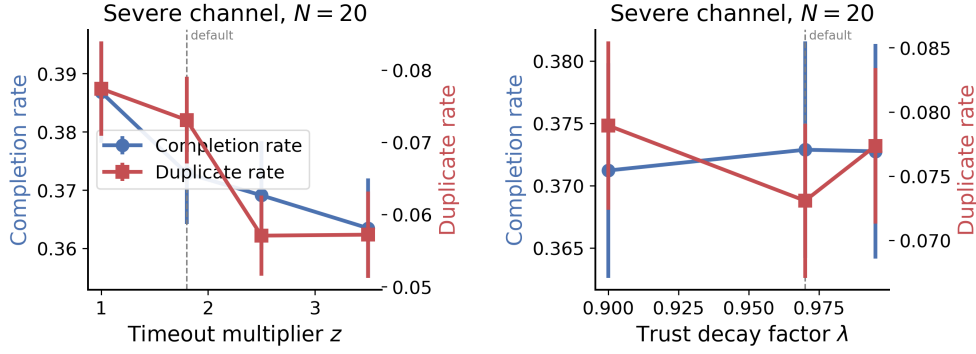


Fig. 10 Sensitivity of completion rate and duplicate rate to TGPR’s timeout multiplier z (left) and trust decay factor λ (right). z controls a real, monotonic trade-off between completion rate and duplicate/messaging efficiency; λ has little effect in the range tested.

Completion rate across channel-quality scenarios

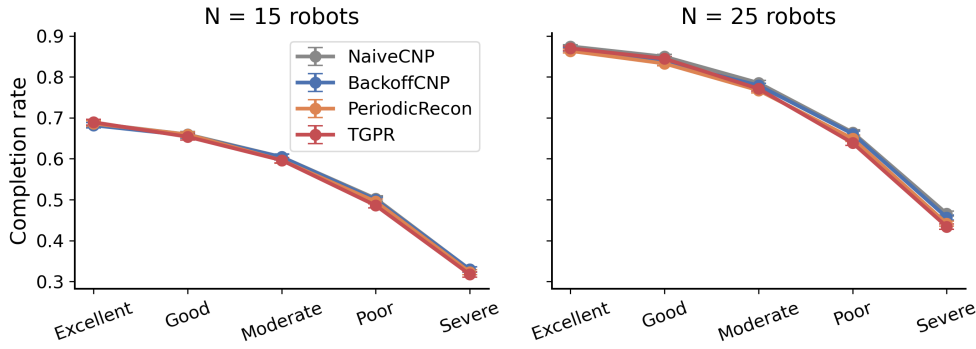


Fig. 11 Completion rate across the five channel-quality scenarios at both team sizes (mean $\pm$ 95% CI over 480 trials/protocol/condition, except severe/ $N=25$ at 483 trials). All four protocols track closely under good channel conditions and separate as conditions worsen, with TGPR trailing rather than leading.

7 Discussion and Limitations

We designed this study to report what the mechanism actually does, including where it does not help, and several findings are worth stating plainly rather than deferring to a generic future-work paragraph.

The efficiency and duplication advantages are real and appear mechanistically well understood. TGPR’s reductions in messaging overhead and duplicate

Duplicate-execution rate across channel-quality scenarios

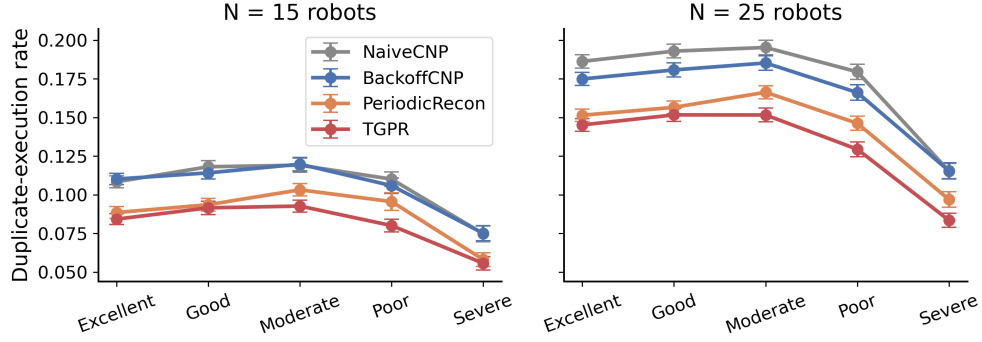


Fig. 12 Duplicate-execution rate across channel scenarios and team sizes. TGPR is the lowest-duplicate protocol in every condition tested.

Messages per completed task across channel-quality scenarios

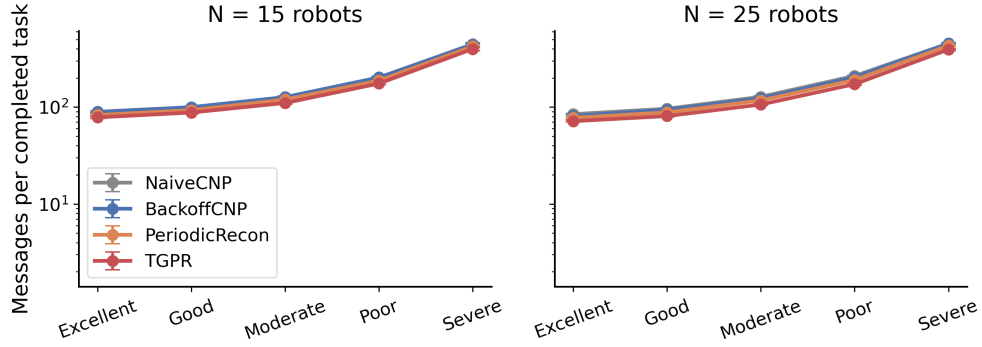


Fig. 13 Messages sent per completed task (log scale) across channel scenarios and team sizes. TGPR's efficiency advantage over all three baselines grows as channel quality worsens.

execution are large, consistent across every channel condition and team size we tested, and grow precisely where they matter most: as channel quality worsens or more of the fleet is hardware-degraded. This is consistent with the mechanism's design – trust-discounted bidding steers awards away from robots whose award-acknowledgment round trip has recently failed, the adaptive timeout avoids the fixed baselines' one-size-fits-all wait, and the confidence-gated bystander claim cuts a full round trip out of the recovery path when it succeeds. For deployments where bandwidth is scarce or duplicated physical effort is costly (e.g., battery-limited field robots), this is a substantial and, we believe, genuine improvement.

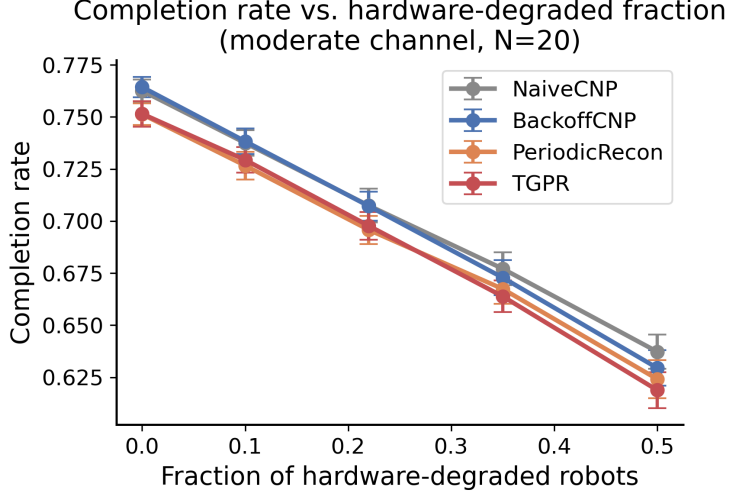


Fig. 14 Completion rate vs. fraction of hardware-degraded robots (moderate channel, $N = 20$, 400 trials/protocol/fraction). All protocols decline together as more of the fleet becomes genuinely unreliable; TGPR’s small completion-rate deficit relative to NaiveCNP and BackoffCNP persists, and widens slightly, across the full range.

The completion-rate cost is real, and our ablation contradicts our own hypothesis about it. We expected that if TGPR paid a completion-rate cost, that cost would shrink as the fraction of genuinely unreliable robots increased, on the reasoning that the trust mechanism would increasingly earn back its overhead by correctly avoiding bad hardware. We observed the opposite: the completion-rate gap between TGPR and the naive baseline widens from 1.1 to 1.9 percentage points as the degraded-unit fraction rises from 0% to 50%. We traced this to a specific, confirmed mechanism rather than leaving it as speculation. First, the bystander confirmation grace window (Section 4) delays the coordinator’s own fallback reassignment by a fixed number of ticks on every recovery cycle, whether or not a bystander is available to preempt it; under frequent failures this fixed tax accumulates. Second, and confirmed directly by the trust trace in Figure 9, the timeout-update rule inflates a robot’s delay-variance estimate on every observed failure (Section 4) without a compensating decay when failures stop, so under sustained high failure rates – exactly the severe-channel and high-degraded-fraction conditions where the effect is largest – learned timeouts ratchet upward over the course of a mission toward the timeout cap rather than stabilizing, and the gap between healthy and degraded robots’ learned trust narrows rather than widens. This also offers the most direct explanation for why we did not observe the hypothesized within-mission learning curve: rather than converging to a stable, informative estimate that improves decisions over time, the estimator’s variance term drifts upward under sustained stress, which shows up as the mild late-mission decline in Section 6. The parameter sensitivity analysis (Figure 10) confirms this is a real, controllable trade-off rather than an unexplained artifact: reducing the timeout multiplier z recovers up to 1.4 percentage points of completion rate at a small cost to duplicate rate and messaging efficiency, while the trust decay factor λ has negligible

effect, correctly implicating timeout sizing rather than trust forgetting as the primary lever. A bounded or periodically-decaying variance term, or decoupling the bystander grace window from the coordinator’s own recovery latency, are natural next steps we did not have space to fully evaluate here.

Trust requires a persistent, learnable signal to exploit. Our first simulation design – distance- and transient-jamming-dependent reliability only, with no persistent per-robot hardware term – produced no measurable advantage for TGPR over the simpler baselines on any metric, because there was no stable per-robot property left to learn once distance and transient interference were already implicitly available to every protocol through its own local observations. We view this as a substantive finding in its own right, independent of the completion-rate result above: trust-based mechanisms of this kind should not be expected to help in fleets whose unreliability is purely positional and transient, and claims of benefit should be checked against an ablation of exactly this kind before being taken at face value.

Star-topology simplification. We modeled communication as a coordinator-centric star topology rather than a full peer-to-peer mesh with multi-hop relaying. This is a reasonable approximation for fleets that rely on a fixed base station or relay (common in warehouse and controlled-facility deployments) but understates the coordination problem in fully ad hoc mesh deployments, where trust would need to be estimated and shared per hop rather than per direct link.

Combined rather than directional trust. For tractability we learn a single combined round-trip reliability estimate per robot rather than separately disentangling downlink and uplink reliability, even though our channel model itself includes an uplink/downlink asymmetry. A directionally disentangled estimator could in principle react faster to a fault that affects only one direction of a robot’s radio, at the cost of requiring more observations to reach the same statistical confidence.

Ground-truth task completion. We assumed, as is common in this literature, that physical task completion is reliably observable by an external means (a supervisory system or sensor) independent of the allocation handshake under study. This is a deliberate scoping decision to isolate the reliability of the auction protocol itself; a deployment in which completion reporting is *also* unreliable would compound both problems and is outside the scope of this paper.

Simulation-only validation. All results reported here come from a discrete-event simulation with a parameterized, but ultimately synthetic, channel model. We calibrated channel and jamming parameters to be qualitatively consistent with reported behavior of lossy 802.11-style links in the multi-robot communication literature [3, 12], but we have not validated TGPR on physical radios or a hardware-in-the-loop testbed, and we do not claim the specific numeric results transfer unchanged to a different channel model or deployment. We regard this as the most important limitation of the present work rather than a minor caveat: simulated packet loss and a real radio’s fading, multipath, and MAC-layer retry behavior can differ in ways that only a hardware or ROS 2 testbed would reveal, and a small-scale validation – e.g., a handful of ground robots on ROS 2 with artificially degraded Wi-Fi on a subset of units, reproducing the hardware-heterogeneity ablation of Section 6 at a scale of $N=4-6$ – is the concrete next step needed before TGPR’s numeric advantages (and cost) should be treated as

anything more than simulation evidence. We did not have the physical robot access to conduct this within the scope of the present paper and report this as an open validation gap rather than an implied equivalence to a hardware result.

8 Conclusion

We introduced Trust-Gated Predictive Reallocation (TGPR), a decentralized auction mechanism for multi-robot task allocation that treats per-robot communication reliability as a first-class, Bayesian-estimated quantity rather than an undifferentiated source of noise to be absorbed by fixed retries. Across a systematic sweep of five channel-quality regimes, two team sizes, and a targeted hardware-heterogeneity ablation totaling 480 trials per main-sweep condition and 400 per ablation condition, TGPR delivers a substantial, statistically robust reduction in duplicate-execution waste (23.9% relative) and per-completed-task communication overhead (13.6% relative), with both advantages growing under the harsher channel and higher-heterogeneity conditions where they matter most. It does not, however, improve overall task completion rate, and in poor and severe channel conditions it completes significantly fewer tasks than the simplest fixed-timeout baseline – a gap our ablation shows widening, not narrowing, as the fleet’s hardware heterogeneity increases, contrary to our initial hypothesis. We attribute this cost primarily to unbounded variance inflation in the adaptive-timeout estimator under sustained failures and, secondarily, to a fixed latency tax from the bystander-claim grace window, and we view fixing this coupling as the clearest next step for this line of work. We report both the advantage and the cost as we found them: TGPR is a substantial win for deployments where communication bandwidth or duplicated physical effort is the binding constraint, but it is not a strict improvement over simpler auctions when raw task-completion throughput is the only metric that matters, and we think that operating boundary is at least as useful to the field as the cases where the mechanism wins. Figure 15 summarizes this trade-off in one view.

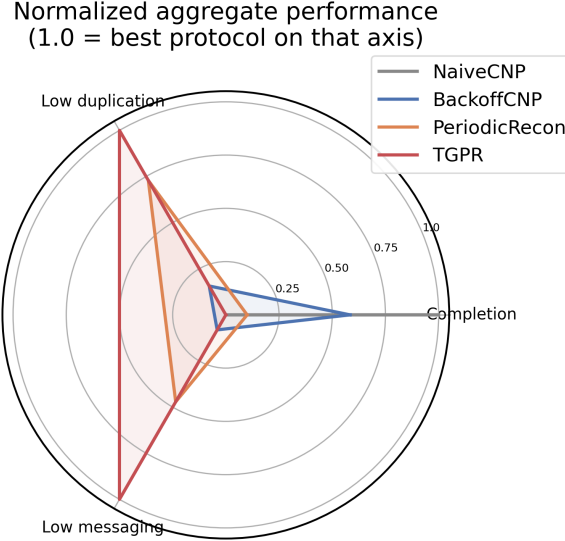


Fig. 15 Normalized aggregate comparison across the three headline metrics (1.0 = best protocol on that axis, averaged over all main-sweep conditions). TGPR dominates on duplicate rate and messaging efficiency and trails slightly on completion rate – the trade-off this paper’s title claims and Section 6 quantifies.

Declarations

- **Funding:** This research received no external funding.
- **Competing Interests:** The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Ethics Approval and Consent to Participate:** This study did not involve human participants, human data, or animals. Ethical approval and informed consent were therefore not required.
- **Consent for Publication:** Not applicable.
- **Data Availability:** No publicly available datasets were used in this study. The experiments were conducted using a custom simulation environment developed by the author. The full simulation environment, experiment and analysis code, per-seed simulation logs, and raw per-episode outcome arrays underlying every figure and table in this manuscript are publicly available at <https://github.com/hasib1233333/Trust-Gated-Predictive-Reallocation>.
- **Code Availability:** See Data Availability above.
- **Author Contributions:** M.H. conceived the trust-gated reallocation mechanism, designed and implemented the simulation testbed and all compared protocols, ran the experiments, performed the statistical analysis, and wrote the manuscript.

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